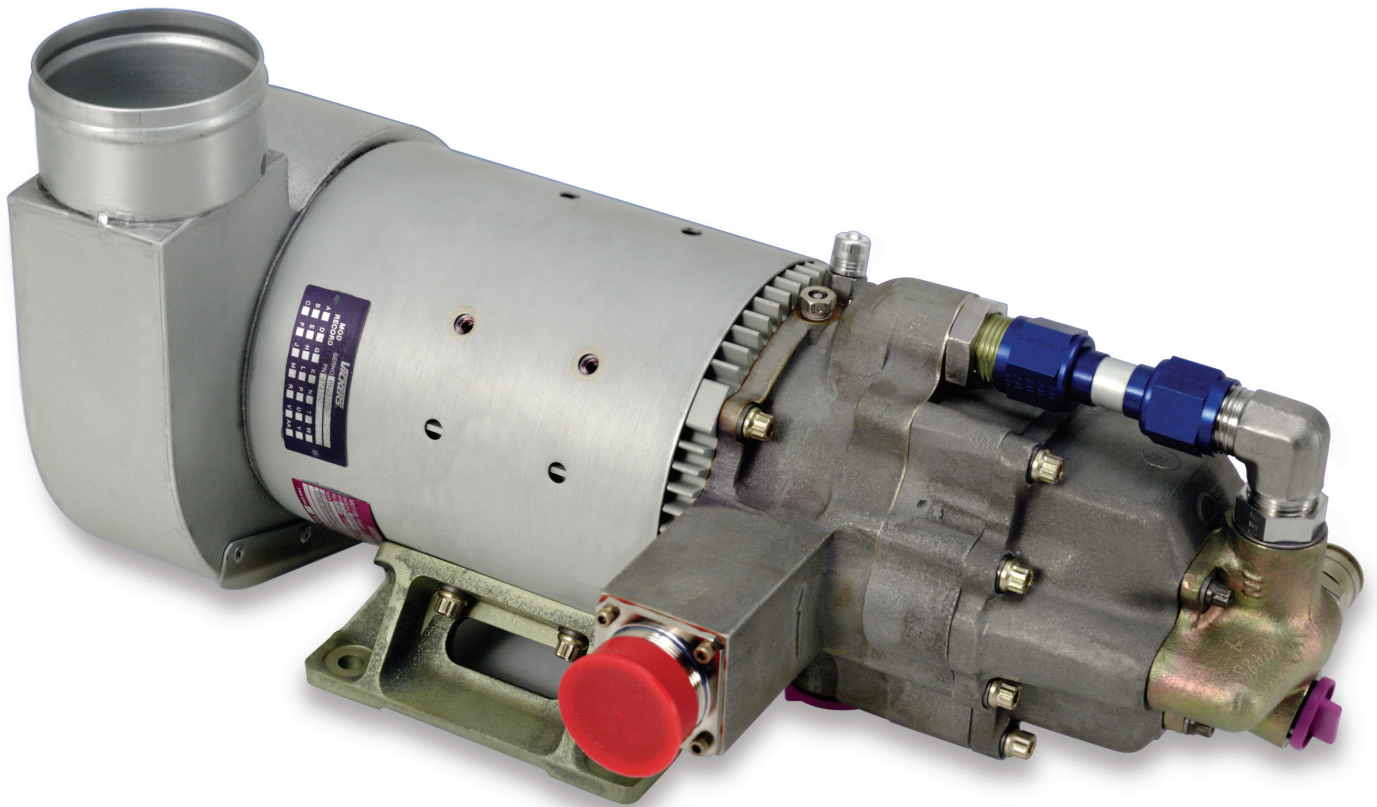




Eaton's Vickers product line includes the MPEV3-032-15 AC electric motor-driven hydraulic pump, designed and manufactured to meet Airbus' overall design criteria for reliability and high maintainability in every area of the airframe and system design.

The MPEV3-032-15 was developed for the Airbus A320, A330 and A340 series aircraft. These aircraft have proven to be among the most advanced and efficient commercial aircraft flying today.

The MPEV3-032-15 motorpump provides hydraulic power for in-flight and system ground operation and helps Airbus and the airlines meet their objectives of lower operational costs and higher reliability.

The MPEV3-032-15 electric motorpump is a high-performance, low-noise unit designed to provide customers with long life and low operating costs.

The motorpump consists of an AC motor integrated with a variable displacement inline hydraulic pump. A fan provides cooling for the electric motor.

An inlet boost pump is included in the motorpump providing super-charge capabilities to the inline pump in the event that the reservoir pressure drops to atmospheric pressure. The boost pump also supplies additional flow to cool the electric motor and tail bearing.

Electric Motor

The air/fluid cooled electric motor is a 7600 rpm, 400 Hz induction-type motor featuring rugged squirrel cage rotor and star wound stator, which converts the electrical energy to rotary shaft motion to drive the boost pump, cooling fan, and inline piston pump.

The electric motor housing serves as a cooling jacket for the motor by providing a flow path to direct air flow across the stator cooling fins.

Hydraulic Pump

The hydraulic pump is a basic 0.26 cu.in./rev/ (4.26 ml/rev) variable displacement unit designed for continuous operation in a 3000 psi (207 bar) aircraft hydraulic system. The seven piston pump is designed to provide 8.5 gpm (32 L/min) flow without compromising the integrity of the compact rotating group.

Pump Controls

The pump is supplied with a "constant horsepower" compensator which limits power output to a given value and maintains it at nearly a constant level within a given flow range. It has the advantage of keeping electrical power consumption requirements to a minimum.

The purpose of the constant horsepower compensator is to begin flow reduction at a given intermediate system pressure (point B), and to cause the pump to reach rated pressure at a second intermediate flow (point C).

In normal operation if there is no resistance to flow, this will register zero system pressure and maximum flow (yoke at maximum angle). As the load is increased, the pressure rises, and system flow remains maximum until the pressure reaches point "B" (2150 psi/150 bar) setting of the compensator valve spring. This pressure is then sufficient to center the compensator valve spool.

As the load increases further, causing the system pressure to exceed 2150 psi/150 bar, the compensator valve spool is moved, and control flow from the discharge package is metered to the yoke actuator piston.

When the pressure on the pistons exceeds the force holding the yoke in the full flow position, the yoke will move to a shallower angle, thereby reducing the flow. As the yoke moves, it compresses the yoke spring opposing the load applied by the yoke actuator piston. When the yoke spring load equals the opposing force (control pressure times the actuator piston area), the compensator spool moves

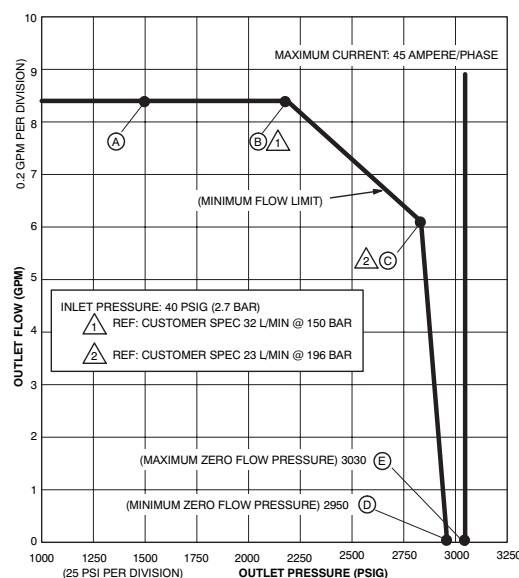
back to the neutral (mid) position, and the new yoke position is maintained.

At point "C" (2850 psi/196 bar), the yoke spring is compressed to the maximum allowable, and the compensator operates as a standard "Flat cut-off" type compensator. If the load is decreased, the system pressure is temporarily reduced, and the compensated valve spool will open the yoke actuator piston chamber to case pressure. The yoke return spring will cause the yoke angle to increase until discharge pressure and another spring forces it to balance again.

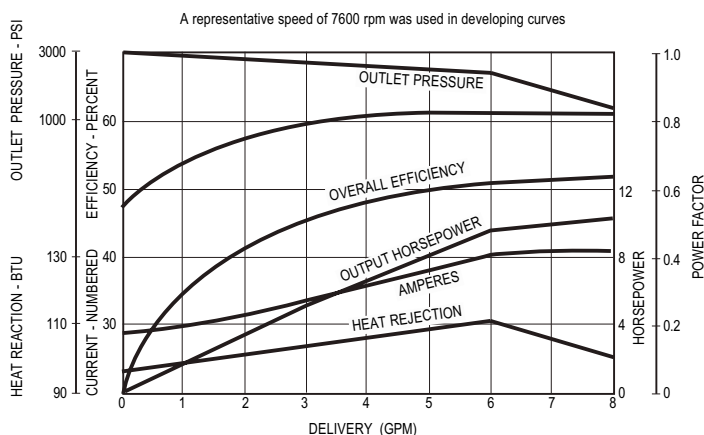
Features and Benefits

- New Replaceable Stator
- Temperature Switches
- Rugged Squirrel Cage Rotor
- Simplified Design
- Increased Bearing Load
- Growth Potential
- Improved Efficiency
- Lower Consumer Current
- Lower Operating Temperature

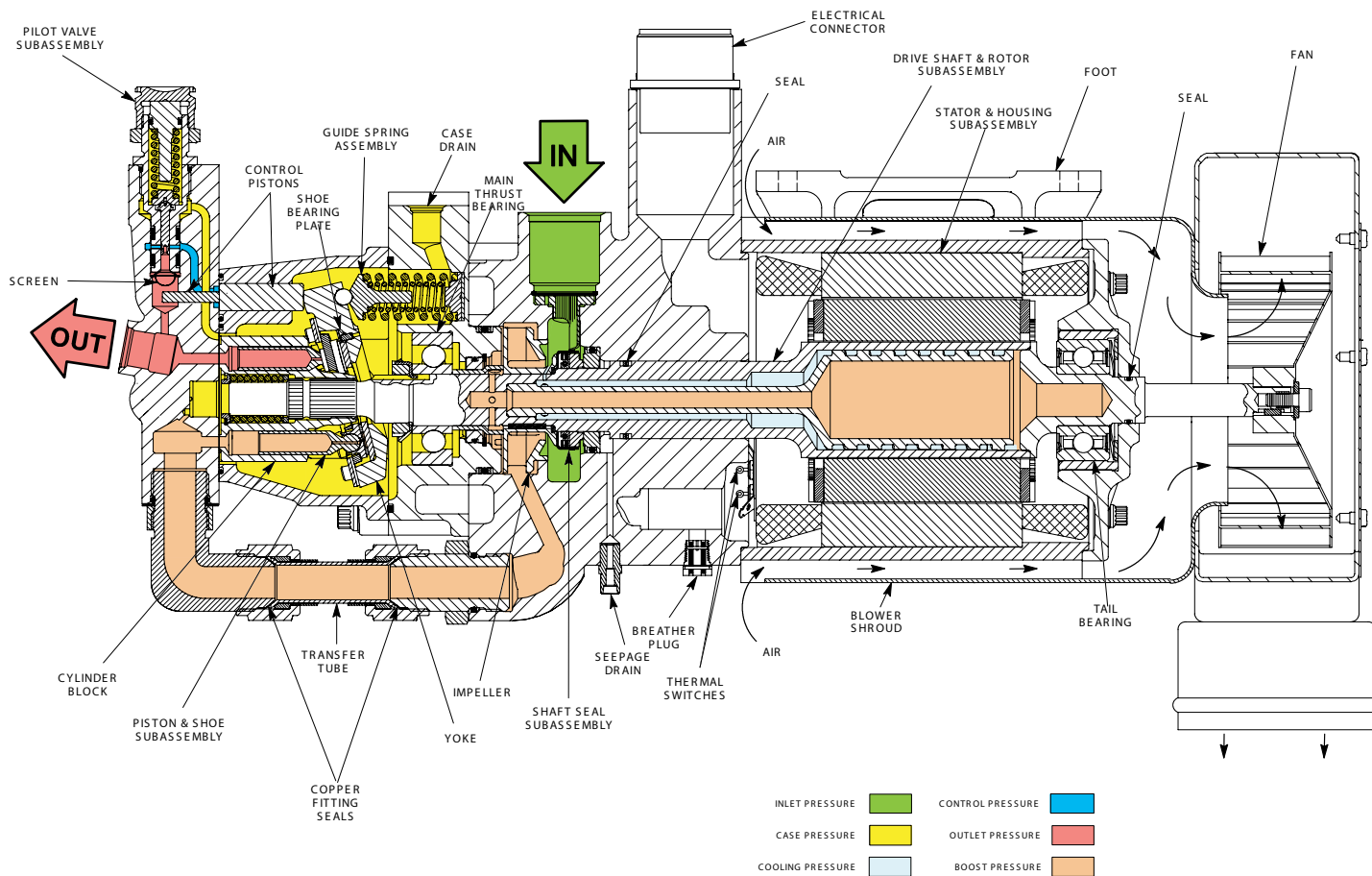
Motorpump Performance



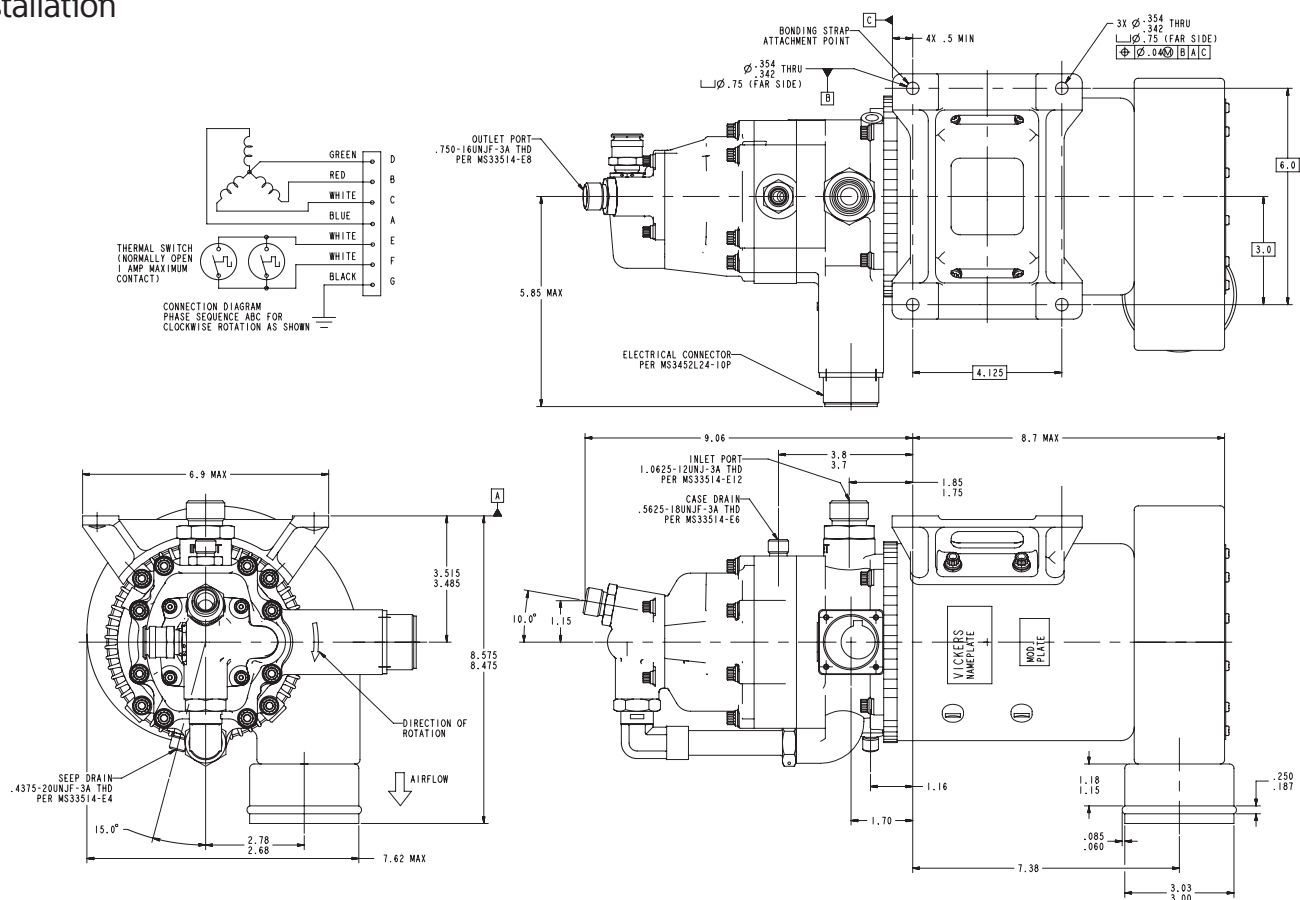
Pump Characteristics



Diagram



Installation



Eaton
Aerospace Group
Fuel & Motion Control Systems Division
5353 Highland Drive
Jackson, Mississippi 39206-3449
Phone: (601) 981-2811
Fax: (601) 987-5255